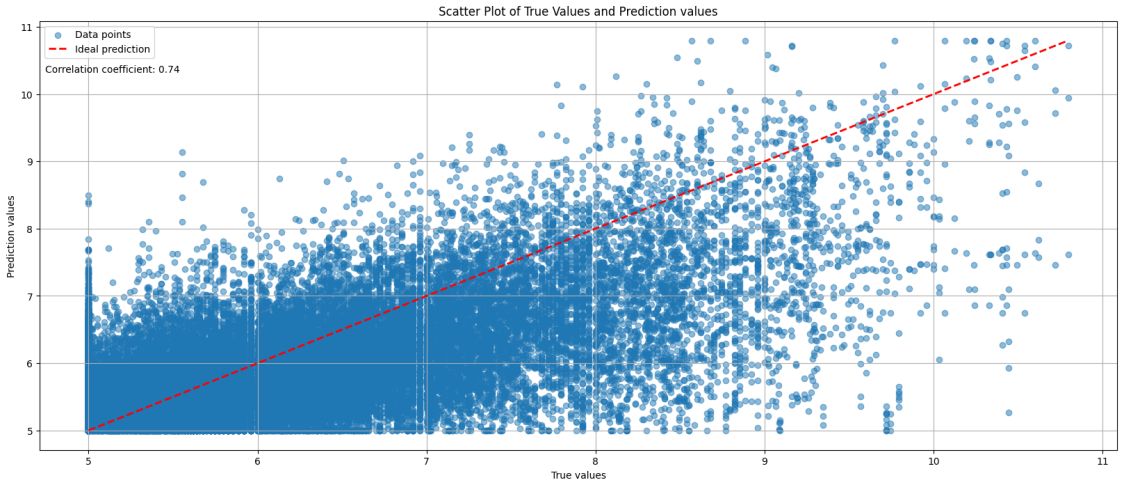
statistics

November 5, 2026

[2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import json
import re
```

[ ]:
```
# Scatter plot of true values and prediction values
file_path = '/Users/mikkohapponen/Documents/Tiedostot/Opiskelu/DI/Diplomityö/
 ↪Data/Models/Results/predictions_davis.csv'
df = pd.read_csv(file_path)
plt.figure(figsize=(20, 8))
plt.scatter(df['True values'], df['Predictions'], alpha=0.5, label='Data␣
 ↪points')
plt.plot([df['True values'].min(), df['True values'].max()],
         [df['True values'].min(), df['True values'].max()],
         color='red', linestyle='dashed', linewidth=2, label='Ideal prediction')
plt.grid(True)
correlation = np.corrcoef(df['True values'], df['Predictions'])[0, 1]
plt.text(0.005, 0.88, f'Correlation coefficient: {correlation:.2f}',
         transform=plt.gca().transAxes, fontsize=10, color='black')
plt.xlabel("True values")
plt.ylabel("Prediction values")
plt.title("Scatter Plot of True Values and Prediction values")
plt.legend()
plt.show()
```



[3]:
```
# Compute mean and standard deviation
file_path = '/Users/mikkohapponen/Documents/Tiedostot/Opiskelu/DI/Diplomityö/
 ↪Data/Models/Results/predictions_davis.csv'
df = pd.read_csv(file_path)
true_mean, true_std = df['True values'].mean(), df['True values'].std()
pred_mean, pred_std = df['Predictions'].mean(), df['Predictions'].std()
print(f"For true values, mean is {true_mean:.3f} and standard deviation is␣
 ↪{true_std:.3f}")
print(f"For prediction values, mean is {pred_mean:.3f} and standard deviation␣
 ↪is {pred_std:.3f}")
```

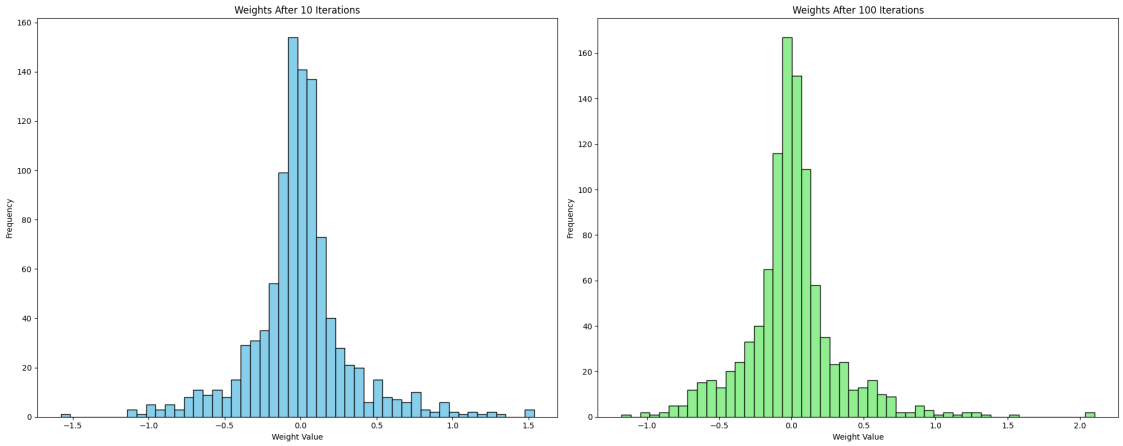
```
For true values, mean is 5.452 and standard deviation is 0.895
For prediction values, mean is 5.489 and standard deviation is 0.674
```

[5]:
```
# Compute mean absolute error, mean squared error, root mean squared error, and␣
 ↪r-squared error
file_path = '/Users/mikkohapponen/Documents/Tiedostot/Opiskelu/DI/Diplomityö/
 ↪Data/Models/Results/predictions_davis.csv'
df = pd.read_csv(file_path)
true_values = df['True values']
predictions = df['Predictions']
mae = mean_absolute_error(true_values, predictions)
mse = mean_squared_error(true_values, predictions)
rmse = np.sqrt(mse)
r2 = r2_score(true_values, predictions)
print(f"MAE is {mae:.3f}")
print(f"MSE is {mse:.3f}")
print(f"RMSE is {rmse:.3f}")
print(f"R-squared is {r2:.3f}")
```

```
MAE is 0.364
```

```
MSE is 0.359
RMSE is 0.599
R-squared is 0.552
```

[ ]:
```python
# Plot histograms of weight distributions for different iterations
with open("weights_davis_both_10iterations.json", "r") as f1:
    data_early = json.load(f1)
with open("weights_davis_both_100iterations.json", "r") as f2:
    data_late = json.load(f2)
w_early = np.array(data_early["weights (w)"])
w_late = np.array(data_late["weights (w)"])
plt.figure(figsize=(20, 8))
plt.subplot(1, 2, 1)
plt.hist(w_early, bins=50, color='skyblue', edgecolor='black')
plt.title("Weights After 10 Iterations")
plt.xlabel("Weight Value")
plt.ylabel("Frequency")
plt.subplot(1, 2, 2)
plt.hist(w_late, bins=50, color='lightgreen', edgecolor='black')
plt.title("Weights After 100 Iterations")
plt.xlabel("Weight Value")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



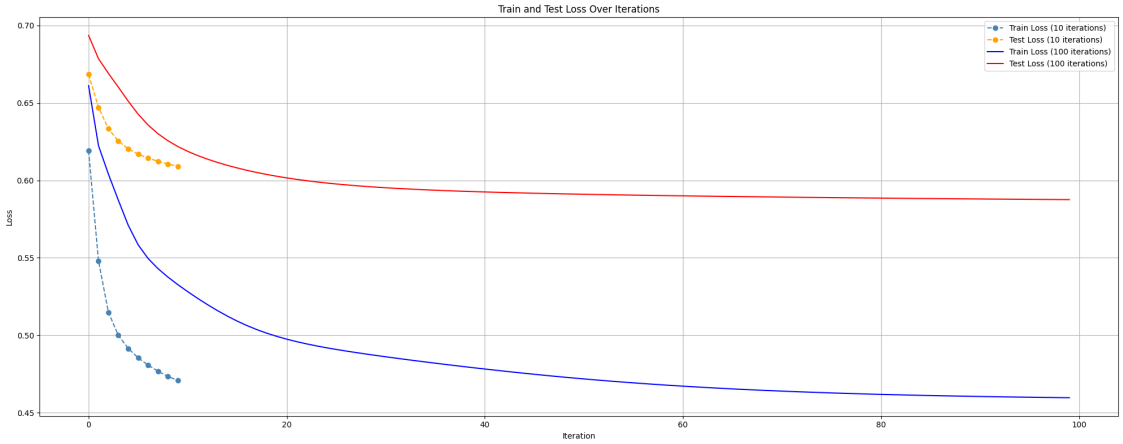
[3]:
```python
# Log outputs
log_10 = """
#Iter=  0        Train=0.619215        Test=0.668475
#Iter=  1        Train=0.548081        Test=0.647162
#Iter=  2        Train=0.514883        Test=0.633396
#Iter=  3        Train=0.500018        Test=0.625399
#Iter=  4        Train=0.491514        Test=0.62043
```

```
#Iter=  5        Train=0.485502        Test=0.616998
#Iter=  6        Train=0.480767        Test=0.614424
#Iter=  7        Train=0.476828        Test=0.612358
#Iter=  8        Train=0.473529        Test=0.610618
#Iter=  9        Train=0.470805        Test=0.609079
"""
log_100 = """
#Iter=  0        Train=0.661148        Test=0.693615
#Iter=  1        Train=0.622431        Test=0.678573
#Iter=  2        Train=0.604079        Test=0.669119
#Iter=  3        Train=0.587291        Test=0.660246
#Iter=  4        Train=0.571058        Test=0.651176
#Iter=  5        Train=0.558456        Test=0.642783
#Iter=  6        Train=0.549646        Test=0.635786
#Iter=  7        Train=0.543082        Test=0.630169
#Iter=  8        Train=0.537629        Test=0.625641
#Iter=  9        Train=0.532753        Test=0.621919
#Iter= 10        Train=0.52821        Test=0.618788
#Iter= 11        Train=0.523898        Test=0.616094
#Iter= 12        Train=0.519825        Test=0.613731
#Iter= 13        Train=0.515989        Test=0.611631
#Iter= 14        Train=0.512433        Test=0.609745
#Iter= 15        Train=0.509179        Test=0.608042
#Iter= 16        Train=0.506251        Test=0.606498
#Iter= 17        Train=0.503646        Test=0.605098
#Iter= 18        Train=0.501343        Test=0.603827
#Iter= 19        Train=0.499309        Test=0.602674
#Iter= 20        Train=0.49751        Test=0.601629
#Iter= 21        Train=0.495907        Test=0.600682
#Iter= 22        Train=0.49447        Test=0.599823
#Iter= 23        Train=0.493169        Test=0.599045
#Iter= 24        Train=0.491978        Test=0.598339
#Iter= 25        Train=0.490875        Test=0.597698
#Iter= 26        Train=0.489839        Test=0.597116
#Iter= 27        Train=0.488856        Test=0.596586
#Iter= 28        Train=0.487918        Test=0.596103
#Iter= 29        Train=0.487014        Test=0.595662
#Iter= 30        Train=0.486129        Test=0.595258
#Iter= 31        Train=0.485269        Test=0.594887
#Iter= 32        Train=0.484425        Test=0.594545
#Iter= 33        Train=0.4836        Test=0.594231
#Iter= 34        Train=0.482791        Test=0.593939
#Iter= 35        Train=0.481997        Test=0.593668
#Iter= 36        Train=0.481218        Test=0.593415
#Iter= 37        Train=0.480453        Test=0.593179
#Iter= 38        Train=0.4797        Test=0.592957
#Iter= 39        Train=0.478961        Test=0.592749
```

```
#Iter= 40       Train=0.478238       Test=0.592553
#Iter= 41       Train=0.477526       Test=0.592368
#Iter= 42       Train=0.476829       Test=0.592193
#Iter= 43       Train=0.476147       Test=0.592027
#Iter= 44       Train=0.475481       Test=0.591869
#Iter= 45       Train=0.474831       Test=0.591718
#Iter= 46       Train=0.474199       Test=0.591575
#Iter= 47       Train=0.473584       Test=0.591437
#Iter= 48       Train=0.472988       Test=0.591305
#Iter= 49       Train=0.472409       Test=0.591179
#Iter= 50       Train=0.471848       Test=0.591057
#Iter= 51       Train=0.471304       Test=0.59094
#Iter= 52       Train=0.470777       Test=0.590826
#Iter= 53       Train=0.470268       Test=0.590717
#Iter= 54       Train=0.469775       Test=0.590611
#Iter= 55       Train=0.469301       Test=0.590508
#Iter= 56       Train=0.468843       Test=0.590409
#Iter= 57       Train=0.468401       Test=0.590312
#Iter= 58       Train=0.467973       Test=0.590218
#Iter= 59       Train=0.467562       Test=0.590126
#Iter= 60       Train=0.467166       Test=0.590036
#Iter= 61       Train=0.466785       Test=0.589948
#Iter= 62       Train=0.466418       Test=0.589863
#Iter= 63       Train=0.466065       Test=0.589779
#Iter= 64       Train=0.465727       Test=0.589697
#Iter= 65       Train=0.465401       Test=0.589617
#Iter= 66       Train=0.465089       Test=0.589538
#Iter= 67       Train=0.464789       Test=0.589461
#Iter= 68       Train=0.464501       Test=0.589385
#Iter= 69       Train=0.464226       Test=0.589311
#Iter= 70       Train=0.463962       Test=0.589238
#Iter= 71       Train=0.463709       Test=0.589166
#Iter= 72       Train=0.463467       Test=0.589095
#Iter= 73       Train=0.463235       Test=0.589026
#Iter= 74       Train=0.463013       Test=0.588957
#Iter= 75       Train=0.462799       Test=0.58889
#Iter= 76       Train=0.462593       Test=0.588824
#Iter= 77       Train=0.462397       Test=0.588759
#Iter= 78       Train=0.462209       Test=0.588695
#Iter= 79       Train=0.462029       Test=0.588632
#Iter= 80       Train=0.461857       Test=0.58857
#Iter= 81       Train=0.461693       Test=0.58851
#Iter= 82       Train=0.461536       Test=0.58845
#Iter= 83       Train=0.461386       Test=0.588392
#Iter= 84       Train=0.461242       Test=0.588335
#Iter= 85       Train=0.461105       Test=0.588279
#Iter= 86       Train=0.460974       Test=0.588223
```

```
#Iter= 87        Train=0.460849        Test=0.588169
#Iter= 88        Train=0.460729        Test=0.588116
#Iter= 89        Train=0.460615        Test=0.588064
#Iter= 90        Train=0.460505        Test=0.588013
#Iter= 91        Train=0.4604        Test=0.587963
#Iter= 92        Train=0.460298        Test=0.587914
#Iter= 93        Train=0.4602        Test=0.587866
#Iter= 94        Train=0.460106        Test=0.587819
#Iter= 95        Train=0.460017        Test=0.587774
#Iter= 96        Train=0.459931        Test=0.587729
#Iter= 97        Train=0.459848        Test=0.587685
#Iter= 98        Train=0.459769        Test=0.587642
#Iter= 99        Train=0.459693        Test=0.5876
"""

# Plot train and test loss curves for different iterations
def parse_log(log_str):
    train, test = [], []
    for line in log_str.strip().splitlines():
        match = re.search(r'Train=([\d.]+)\s+Test=([\d.]+)', line)
        if match:
            train.append(float(match.group(1)))
            test.append(float(match.group(2)))
    return train, test
train_loss_10, test_loss_10 = parse_log(log_10)
train_loss_100, test_loss_100 = parse_log(log_100)
plt.figure(figsize=(20, 8))
plt.plot(train_loss_10, label='Train Loss (10 iterations)', marker='o',␣
 ↪linestyle='--', color='steelblue')
plt.plot(test_loss_10, label='Test Loss (10 iterations)', marker='o',␣
 ↪linestyle='--', color='orange')
plt.plot(train_loss_100, label='Train Loss (100 iterations)', color='blue')
plt.plot(test_loss_100, label='Test Loss (100 iterations)', color='red')
plt.xlabel("Iteration")
plt.ylabel("Loss")
plt.title("Train and Test Loss Over Iterations")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```


Train and Test Loss Over Iterations
Train Loss (10 iterations)
Test Loss (10 iterations)
Train Loss (100 iterations)
Test Loss (100 iterations)
Loss
Iteration
0.70
0.65
0.60
0.55
0.50
0.45
0
20
40
60
80
100